

CSC458H1: Computer Networks — Fall 2026

Worksheet 1: Layering, Protocols, and Encapsulation

Q0 (Knowledge Check) For each statement below, specify if it is true or false:

0.a How does Manchester Encoding make clock recovery easier?

- ☐ By sending a sequence of alternating 0s and 1s before the main frame.
- ☐ By sending a sequence of 1s at the beginning of each frame.
- ☐ By embedding the clock cycle in the signal.
- ☐ By sending the clock signal as a separate signal.

0.b Which encoding scheme do you suggest in order to provide higher bandwidth?

- ☐ Manchester
- ☐ 4B/5B

0.c What is the maximum overhead in byte-stuffing algorithm (excluding the flags marking the start and end of the frame)?

- ☐ 25%
- ☐ 50%
- ☐ 100%
- ☐ 125%

0.d Fill out the blank in the following statement. Justify your answer.

In 4B/5B Encoding that we discussed in the lecture, which uses NRZI encoding as part of its procedure, a signal transition will occur at least every ____ bit times (i.e., clock cycle).

0.e (Applications with Multiple Streams)

0.e.i Name three applications that require tying multiple streams together. For each application, explain what each stream corresponds to.

0.e.ii In the context of OSI model, which layer manages tying multiple streams together?

0.e.iii In the context of 4-layer Internet model, which layer manages tying multiple streams together?

0.f (Transport Layer and Multiplexing) Which of the following statements are true?

- ☐ Transport layer multiplexing assigns different IP addresses to multiple data streams.
- ☐ TCP multiplexing manages multiple logical sessions over a single application-level connection, such as synchronizing audio and video streams in a video call.
- ☐ Transport layer multiplexing ensures that data from multiple applications on the same device is sent and received correctly using port numbers.
- ☐ Application layer is responsible for splitting data into chunks to be sent over multiple streams.

Q1 (Reviewing Layering and Encapsulation) In Figure 1, Host A starts sending one packet to host B at time t_0 . At each time step, complete the missing sections in Figure 2 to indicate which headers are appended to the payload. For reference, the packet headers at time t_2 are provided. Under the context of OSI reference model, In Figure 1, Hypertext Transfer Protocol (HTTP) operates at L7, Transmission Control Protocol (TCP) operates at L4, Internet Protocol (IP) operates at L3, Ethernet and Optical Transport Network (OTN) operate at L2.

Q2 (A Day In Life of A Packet) Consider the network in figure 3, where Host A sends a TCP connection request to Host B. The connection request packet travels along the path: $A \rightarrow R_1 \rightarrow R_2 \rightarrow B$. Considering a 4-layer Internet model, describe the actions performed by each layer as the packet moves through the layers of Host A, router R_1 , router R_2 , and Host B. Whenever applicable, explain the packet created by each layer.

[NOTE: At this stage of the course, it is challenging to provide a precise description of the changes the packet undergoes. Our current explanation will remain high-level and approximate. There are many details you are not yet familiar with. For now, focus on understanding key concepts like layering, encapsulation, and decapsulation.]

Q3 (Circuit Switching vs Packet Switching) Suppose several users share a 3 Mbps link and each user requires 150 Kbps when transmitting. Assume that the probability that a given user is transmitting at any given point in time is $p = 0.1$. Answer the following questions.

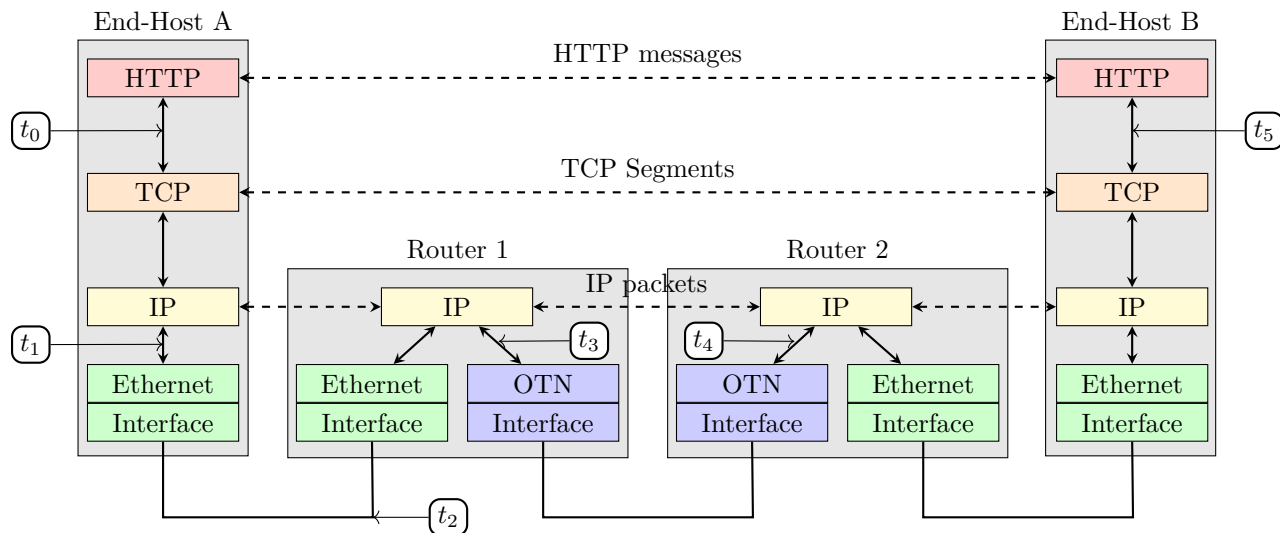


Figure 1: Host A sends one packet to host B at time t_0 .

	L1/L2	L3	L4	L7	
Time = t_0					Payload
Time = t_1					Payload
Time = t_2	Yes	Yes	Yes	Yes	Payload
Time = t_3					Payload
Time = t_4					Payload
Time = t_5					Payload

Figure 2: Which headers are attached to the payload at each time? Put Yes if a layer's header is attached, and No if it is not.

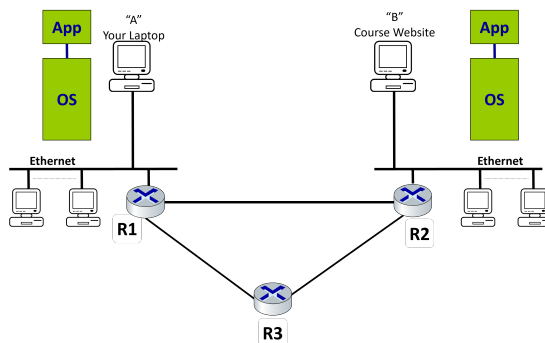


Figure 3: Host A sends a TCP SYN packet to Host B. The packet travels along the path: $A \rightarrow R_1 \rightarrow R_2 \rightarrow B$.

- 3.a When circuit switching is used, how many users can be supported?
- 3.b Suppose there are 120 users. Find the probability that at any given time, exactly n users are

transmitting simultaneously.

- 3.c Find the probability that there are 21 or more users transmitting simultaneously. How do you interpret the meaning of this probability?

Q4 (Statistical Multiplexing) Consider three flows F_1, F_2 , and F_3 sending packets over a link. The sending pattern of each flow is described by how many packets it sends within each one-second interval. Table 1 shows their pattern for the first ten intervals.

Time (s)	1	2	3	4	5	6	7	8	9	10
F_1	1	8	3	15	2	1	1	34	3	4
F_2	6	2	5	5	7	40	21	3	34	5
F_3	45	34	15	5	7	9	21	5	3	34

Table 1: Sending pattern of F_1, F_2 , and F_3 .

- 4.a What is the peak rate of F_1, F_2 , and F_3 ? What is the sum of the peak rates?
- 4.b Now consider all packets to be in the same aggregate flow. What is the peak rate of this aggregate flow?
- 4.c Which is higher - the sum of the peaks, or the peak of the aggregate? Based on the flow patterns, find the statistical multiplexing gain.

Q5 (Statistical Multiplexing Analysis) This question is designed for students with a strong interest in theory. It provides insight into how to determine an adequate buffer size and understand statistical multiplexing gain in a more realistic scenario.

Consider N traffic flows. The arrivals from each flow are governed by a random variable X with a geometric distribution. Specifically, in any time slot, the probability that the number of arrivals from a single flow are equal to k kilobytes (kB) is given by

$$p(X = k) = \begin{cases} 0, & k = 0, \\ (1 - p)^{k-1}p, & k > 0, \end{cases} \quad (1)$$

where $0 < p < 1$. Assume that the N flows are independent and identically distributed.

In every time slot, the arrivals are entered into a buffer with a given size. If arrivals to the buffer exceed the buffer size, the excess traffic is lost. The buffered data is removed from the buffer before the start of the next time slot.¹

The objective is to design the size of the buffer such that the probability that a loss occurs does not exceed a given threshold value.

- 5.a Consider that there is one buffer for each flow, i.e., we have N buffers.² Express the total size of all N buffers needed so that the loss probability of each flow does not exceed ϵ .
- 5.b Consider that there is a single buffer that is shared by all N flows.³ Use the Central Limit Theorem (CLT) to derive the buffer size needed so that the loss probability does not exceed ϵ .
[HINT: A random variable X with Geometric distribution as in (1) has $\mathbb{E}[X] = \frac{1}{p}$ and $\text{Var}[X] =$

¹Hence, the buffer is empty at the beginning of each time slot.

²This is similar to circuit switching resource sharing.

³This is similar to packet switching resource sharing.

$$\frac{1-p}{p^2}.$$

[HINT: Let $\Phi(\cdot)$ denote the cumulative distribution function of the standard normal distribution.

To get $1 - \Phi(z) \approx \epsilon$, you may use the approximation $z = \sqrt{|\ln(2\pi\epsilon)|}.$

5.c Consider the previous parts, find the statistical multiplexing gain?